HAO YU

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Research Profile

Experimental materials researcher specialising in vapour-phase growth of low-dimensional oxides and transition-metal dichalcogenides. Combines controlled synthesis, microscopy, Raman/photoluminescence spectroscopy and transport modelling to relate processing conditions to material structure and optical response. Develops Python tools for reproducible spectroscopy analysis, with complementary experience in photonic-device processing, researcher training and technology assessment.

Education

University of Cambridge Oct. 2022 – present
PhD in Engineering; Department of Engineering
Thesis submitted **30 September 2026**; degree pending. Supervisor: Prof. Stephan Hofmann.
Thesis: *Close-space sublimation as a platform for accelerated process discovery in low-dimensional material growth*.

University of Oxford Oct. 2018 – July 2022
MEng in Materials Science, Upper Second-Class Honours (2:1); Trinity College
Final-year project: *Novel 3D nano-manufacturing method for micro-ring resonator non-volatile tuning*.
Supervisors: Prof. Harish Bhaskaran and Dr Nikolaos Farmakidis.

Doctoral Research

Low-dimensional oxide and chalcogenide growth 2022 – present
PhD researcher, Hofmann Group, University of Cambridge

- Designed and conducted close-space sublimation/CVD experiments linking precursor supply, source–substrate geometry and growth conditions to tungsten oxide and sulphide nanostructure formation; co-wrote the first-author *Nanoscale* paper.
- Optimised NaCl-assisted WS_2 growth through controlled precursor, gap, hydrogen and temperature comparisons. Obtained optically active domains approximately 50–100 μm across, with characterisation supporting monolayer formation.
- Designed a close-space growth configuration and combined local COMSOL flow modelling with analytical precursor-flux models and controlled experiments to assess precursor delivery and source/substrate cross-contamination.
- Integrated personal Raman/PL and SEM measurements with collaborator-acquired operando SEM, TEM/EDS and XPS evidence to interpret source evolution, material conversion and deposited structures.
- Recent work includes operando SEM data analysis, post-growth characterisation and discussions of colleagues' experiments involving applied potential, electron-beam exposure and temperature.
- Contributed WS_2 and MoS_2 samples, growth recipes and a Raman spectral-analysis workflow to a collaborative tutorial on imaging spectroscopic ellipsometry of 2D materials.

Research software: RamanPL_2D

- Developed a Python toolkit for Raman/PL preprocessing, baseline correction, Lorentzian and pseudo-Voigt peak fitting, spatial mapping, batch analysis and metadata-aware export.
- Implemented workflows for extracting and comparing fitted peak parameters from single spectra and mapping datasets. Open-source repository; BSD 3-Clause licence.

Publications

Peer-reviewed journal articles

1. **Hao Yu**, Jinfeng Yang, Bernhard Fickl, Shaoliang Guan, Oliver J. Burton, Gwenhivir Wyatt-Moon, Andrew Flewitt, Bernhard C. Bayer and Stephan Hofmann. "A close-space sublimation approach to tungsten oxide and sulfide nanostructure formation." *Nanoscale* **17**(36) (2025), 21070–21082. doi: 10.1039/D5NR01458A.
   First author; CSS growth, Raman/PL measurements and manuscript co-writing.

2. Sunvir Sahota, Irina Chircă, Oliver J. Burton, **Hao Yu**, Max Rimmer, Jinfeng Yang, Kyungseo Park, Arthur Summers, Siddika Mertdinc-Ulkuseven, Matthew Lindley, Sarah J. Haigh and Stephan Hofmann. "Se nanowire crystal formation via oxidation of 2D $HfSe_2$: A solid-state, in situ reaction coupling for heterogeneous integration technologies." *ACS Applied Nano Materials* **8**(15) (2025), 7608–7615. doi: 10.1021/acsanm.5c00308.
   Contribution: Raman and SEM data acquisition and analysis.

3. Nikolaos Farmakidis, **Hao Yu**, June Sang Lee, Johannes Feldmann, Mengyun Wang, Yuhan He, Samarth Aggarwal, Bowei Dong, Wolfram H. P. Pernice and Harish Bhaskaran. "Scalable high-precision trimming of photonic resonances by polymer exposure to energetic beams." *Nano Letters* **23**(11) (2023), 4800–4806. doi: 10.1021/acs.nanolett.3c00220.
   Second author; measurements, design and simulations. Associated correction: *Nano Letters* **24**(41) (2024), 13103–13104; doi: 10.1021/acs.nanolett.4c03363.

4. Minfei Jian, Jiarui Niu, Wenhua Li, Yuyue Huang, **Hao Yu**, Zheng Lai, Shuli Liu and Elvis Genbo Xu. "How do microplastics adsorb metals? A preliminary study under simulated wetland conditions." *Chemosphere* **309**, Part 1 (2022), 136547. doi: 10.1016/j.chemosphere.2022.136547.
   Contribution: data analysis and manuscript revision.

Submitted tutorial

Ludwig Maximilian Hanauske et al., including **Hao Yu**. "Imaging Spectroscopic Ellipsometry of Van der Waals Quantum Materials: A Practical Tutorial." Submitted to *Materials for Quantum Technology* (2026).
Contribution: WS_2/MoS_2 samples, growth recipes and Raman spectral-analysis workflow.

Manuscript under review

Jinfeng Yang, Ryo Mizuta, Ye Fan, **Hao Yu**, Greg Chu, Sicheng Li, Shaoliang Guan, Xiaofan Lin, Jack Alexander Webber and Stephan Hofmann. "Closed-loop crystal layer writing of atomically thin semiconductor." Under review at *ACS Nano*.
Contribution: Raman/PL measurements and data analysis.

Manuscript in preparation

Hao Yu et al. "Resolving source–transport–growth relationships in salt-assisted close-space CVD of monolayer WS_2." First-author manuscript in preparation; working title.

Selected Poster Presentations

- **2D TMDs 2026**, National University of Singapore, Singapore, 28 June–2 July 2026. *Flux-Resolved Growth Exploration in Close-Space CVD: From WO_{3-x}–WS_2 Core–Shell Nanotubes to WS_2–MoS_2 Heterostructures*.
- **IWEPNM 2026**, Kirchberg in Tirol, Austria; poster session, 1 March 2026. *Direct Fabrication of WS_2–WO_{3-x} Nanotube Devices on Si Substrate via a Close-space Sublimation Approach*.

Further Research Experience

Collaborative nanomaterials characterisation 2025
University of Cambridge

- **MoS_2 crystal-layer writing (June–August):** performed Raman/PL measurements and mapping of directly written MoS_2, assessing layer-dependent optical response and spatial heterogeneity for the manuscript under review.
- **Se nanowire formation (January–April):** acquired Raman and SEM data for the published study; adjusted SEM conditions to manage charging on glass and sapphire and worked with the collaborator to relocate optical/Raman regions of interest.

Photonic-device processing and optical characterisation Sept. 2021 – June 2022
MEng project, University of Oxford

- Processed PMMA-coated micro-ring resonators using chip cleaning, spin coating, baking and UV mask-aligner exposure; linked processing conditions to optical transmission measurements.
- Aligned fibre-to-chip coupling and optimised polarisation for optical transmission measurements in the Oxford micro-ring resonator project.
- Used MATLAB and Lumerical analysis/simulation to interpret resonance changes; contributed measurements, design and simulations to the *Nano Letters* publication.
- Gained supervised RIE experience using an established recipe; electron-beam patterning for the project was performed by a collaborating researcher.

Earlier research: Er–Si nanowire growth, Zhejiang University (June–September 2019): investigated nanowire formation from evaporated Er films using electron-beam evaporation and XRD-based structural analysis. Latch-circuit radiation-hardening project, Nanyang Technological University (May–December 2016): Cadence-based simulation; project received a Bronze Medal at SSEF 2017.

Technical Expertise

Synthesis and processing: Close-space sublimation/CVD; low- and ambient-pressure growth; precursor preparation, oxide sulphidation and controlled process comparisons; PMMA processing and UV mask alignment.

Microscopy and spectroscopy: Independent SEM and SEM–EDS acquisition/analysis; Raman/PL spectroscopy and mapping; Oxford Instruments AZtec/INCA experience; fast-scan SEM acquisition and repeated-frame averaging.

Structural characterisation: Bruker XRD acquisition for WO_3 and suboxides, reference-pattern matching, phase identification and lattice-parameter analysis. Interpretation of collaborator-acquired XPS and TEM/EDS data; AFM and ellipsometry experience/training.

Scientific computing and modelling: Python (NumPy, SciPy, pandas, Matplotlib); MATLAB; COMSOL gas-flow/transport modelling; analytical precursor-flux models; Lumerical photonics simulation.

Teaching and Laboratory Responsibilities

- Trained approximately two new PhD researchers in SEM operation over roughly two years (2024–2026), using demonstrations, practical sessions and observation of their operation; demonstrated SEM imaging of 2D MoS_2 to two NanoDTC cohorts.
- Authored a group SOP for Raman characterisation and analysis and trained researchers in Raman workflows; provided practical guidance in EDS acquisition and cleanroom techniques.
- Maintained low- and ambient-pressure CVD reactors, checked operating pressures and examined gas delivery using mass-flow controllers and needle valves; documented SEM pressure faults and liaised with technical support.
- Prepared and submitted departmental laboratory safety assessments, including CVD chemical handling and laser-heating hazards.

Technology Assessment and Industry Engagement

InnoAngel Fund, Beijing June – August 2026
Investment Analyst Intern; Little Angel Programme

- Produced a compute-in-memory chip sector study covering technical routes, value chains, application scenarios, benchmark companies and startup positioning.
- Conducted founder/team interviews, sector research and preliminary due-diligence assessment across deep-tech fields, including AI infrastructure, robotics, fusion and advanced manufacturing.

Cambridge Judge Business School January – April 2026
EnterpriseTECH Analyst

- Conducted market sizing, application analysis and competitor mapping for photonic photovoltaic technology targeting indoor solar and ambient IoT markets.
- Supported a proposed TRL 2-to-4 roadmap and IP-licensing strategy, including minimum viable product definition, scale-up considerations and revenue-model analysis.

BOE Technology Group January 2023 – December 2025
School Ambassador; UK university engagement

- Organised online and in-person university career fairs and identified researchers for potential MicroLED research collaboration.
- Initiated and maintained contact between BOE and a Cambridge-trained researcher in perovskite quantum-dot synthesis, matching research expertise to next-generation display needs.

Academic Service and Leadership

University of Cambridge, Department of Engineering October 2025 – September 2026
Academic Representative (Postgraduate Research)

- Represented postgraduate researchers on departmental committees, contributing student perspectives on teaching and examinations, AI tools, welfare and inclusion; communicated departmental information and decisions to students.

Cambridge Chinese Students and Scholars Association October 2023 – October 2024
Executive Vice-President

- Directed Cambridge Innovation Month and the Innovation and Entrepreneurship Competition; served as General Director of the 2024 Cambridge Chinese New Year Gala.

Oxford Chinese Students and Scholars Association September 2020 – September 2021
Vice-President

- Co-hosted UK-wide entrepreneurship forums and managed the society's WeChat and Facebook communications.

Awards and Further Development

- **Cambridge Philosophical Society Studentship**, 2025–2026.
- **Exhibition Scholar**, Trinity College, University of Oxford, 2020–2022.
- **Singapore Ministry of Education SM1 Scholarship**, 2014–2017; fully funded school-based scholarship.

Further development: LinkedIn Learning certificates: *Machine Learning with Python* (January 2025); *Artificial Intelligence Foundations: Machine Learning* (November 2021).

Languages: English (fluent), Mandarin (native), Japanese (conversational).